Akhil Krishna R.

Postdoctoral Fellow | Indian Institute of Astrophysics | Bengaluru, India

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Research interests

Galaxy structure and evolution; polar-ring and low-surface-brightness galaxies; disc dynamics and stellar bars; multiwavelength observations; star formation; stellar populations; globular-cluster systems; modelling and numerical simulations.

Appointments and education

2025–present | Postdoctoral Fellow, Indian Institute of Astrophysics, Bengaluru.

2020–2025 | Ph.D. Physics, Christ University, Bengaluru. Thesis: Tracing the Evolution of Polar Ring Galaxies through Multi-wavelength Observations and Cosmological Simulations. Supervisor: Dr. Sreeja S. Kartha.

2019–2020 | Research Fellow, Christ University, Bengaluru. UV upturn in early-type galaxies.

2017–2019 | M.Sc. Physics, Christ University, Bengaluru. Thesis: UV Upturn in Early-Type Galaxies.

2014–2017 | B.Sc. Physics, Catholicate College, Mahatma Gandhi University, Kerala.

First-author refereed publications

2025 | Akhil, K. R., et al. Connecting the dots: Tracing the evolutionary pathway of polar ring galaxies in the cases of NGC 3718, NGC 2685, and NGC 4262. PASA 42, e056. doi:10.1017/pasa.2025.24.

2024 | Akhil, K. R., et al. Deciphering the properties of UV upturn galaxies in the Virgo cluster. MNRAS 534, 4063–4071. doi:10.1093/mnras/stae2382.

2024 | Akhil, K. R., Kartha, S. S., and Mathew, B. Beyond the rings: Polar ring galaxy NGC 4262 and its globular cluster system. MNRAS 530, 2907–2918. doi:10.1093/mnras/stae1061.

2024 | Akhil, K. R., et al. DES J024008.08–551047.5: A new member of the polar ring galaxy family. A&A 681, A35. doi:10.1051/0004-6361/202347357.

Selected co-authored refereed publications

2025 | Sruthi, K., et al. Probing the resolved K–S relation in nearby galaxies: Insights from UVIT and ALMA observations. PASA 42, e140. doi:10.1017/pasa.2025.10088.

2025 | Devaraj, A., et al. Non-linear solar EUV-driven sodium release from the lunar surface: A contrast to the linear PSD model. MNRAS 543, 1079–1091. doi:10.1093/mnras/staf1447.

2025 | Arun, R., et al. Polycyclic Aromatic Hydrocarbons in the Circumstellar Medium of Herbig Ae/Be Stars. RAA 25, 095004. doi:10.1088/1674-4527/ade65c.

2024 | Robin, T., et al. The Interaction Jigsaw: Investigating star formation in interacting galaxies. MNRAS 534, 1902–1912. doi:10.1093/mnras/stae2211.

2024 | Ujjwal, K., et al. Disentangling the association of PAH molecules with star formation: Insights from the James Webb Space Telescope and the UltraViolet Imaging Telescope. A&A 684, A71. doi:10.1051/0004-6361/202347620.

2022 | Ezhikode, S. H., et al. Long-term Optical and Gamma-Ray Variability of the Blazar PKS 1222+216. ApJ 939, 76. doi:10.3847/1538-4357/ac9627.

Selected conference presentations and activities

2026: IIA In-House Symposium (talk); 44th Astronomical Society of India Meeting (talk). 2023: APRIM, Japan (talk); International NUVA e-Meeting (talk); 3rd Australia–ESO Galaxy Transformation Meeting (poster). 2022: ASI Meeting, IIT Roorkee (poster). 2020: ASI Meeting, IISER Tirupati (poster).

Teaching, observational programmes and service

Mentored five M.Sc. students; taught introductory astronomy, Python programming and astronomy laboratory classes. Awarded telescope time: AstroSat/UVIT, 3.6-m Devasthal Optical Telescope (ADFOSC), ESO New Technology Telescope (EFOSC2). WALLABY collaboration member; Astronomical Society of India life member. Workshop resource person and conference organizing committee member.

Source: September 2026 academic CV; public edition prepared without phone number or referee contact details.